

THEMATIC ANALYSIS – CODEBOOK

“Emerging Challenges in Threat Modeling for GenAI-Augmented Systems: A View from the Trenches”

THEME	CODE	DESCRIPTION
SYSTEM TYPE	GenAI	Systems based on generative AI (e.g. LLMs, diffusion models).
	Traditional ML	Systems using classical machine learning models (Like for e.g. decision trees, SVMs).
	Hybrid AI	Combination of symbolic AI and ML models.
SYSTEM MODELLING TECHNIQUE	Data Flow Diagram (DFD)	Shows flow of information within the system.
	Unified Modeling Language (UML)	Standardized system modeling notation.
	Software Architecture Modeling	Models processes and workflows.
	Deployment Diagram	Captures hardware/software deployment of AI.
	Sequence Diagram	Focuses on interactions over time.
	Asset Inventories	Lists system assets relevant to security.
	Use Case Scenarios	Captures functional requirements and user interactions.
SUPPORTING ARTIFACTS	Attack Trees	Hierarchical diagrams modeling possible attacks.
	Risk Cases	Scenarios describing risks and their consequences.
	Threat Catalogs	Predefined list of threats (e.g., adversarial ML threats).
TOOL SUPPORT	Full automation	Tool performs complete threat modelling automatically.
	Semi-automated	Tool exists but requires manual input (for e.g., annotations, labeling).
	No tool support	Entirely manual process.
EVALUATION LIMITATIONS	Usability study	Assesses applicability or ease of use.
	Case study	Performance tested on a real-world case.
	Ad-hoc dataset	Tested on synthetic or small dataset.
	Tool benchmark	Compared to other tools on same dataset/test.
	Proof of concept	Only conceptual demonstration, no real evaluation.
	Scalability	Method difficult to apply in large-scale AI systems.
	Performance	Accuracy, precision, or coverage issues.
	Generalizability	Limited to certain AI domains/datasets.
	Manual effort	Heavy reliance on manual steps.
NOVEL CONTRIBUTIONS	LLM-specific threats	Not mentioning threats Like e.g., prompt injection, data leakage, hallucination.
	Acknowledge LLM threats	Explicit consideration of threats unique to LLMs.
	New methodology	Proposes a novel threat modelling methodology.
	New artifacts	Introduces new supporting artifacts or frameworks.
	Domain adaptation	Tailors existing methods to AI/ML-specific contexts.
	General Novel contribution	Any unique contribution not covered above.